Do Now · retrieval

Signed integers & binary arithmetic

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Choose a type

Q2

Keep the leading zeros

Q3

Read an unsigned byte

Q4

Can it fit?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Sign magnitude — The leftmost bit stores the sign. The remaining bits store the positive magnitude. — CODE:
10000101
sign = 1
magnitude = 0000101 = 5
value = −5

Core idea 2

Two’s complement — The most significant bit has a negative place value. — CODE:
11111011
= −128 + 64 + 32 + 16 + 8 + 2 + 1
= −5

Core idea 3

Add using binary carries — CODE:
0 + 0 = 0
0 + 1 = 1
1 + 1 = 10
1 + 1 + 1 = 11 — Keep the fixed number of result bits. A carry beyond the width is discarded.

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Overflow and subtraction — Signed overflow occurs when the mathematical result is outside the representable range. Two same-sign inputs producing an opposite-sign result is a warning sign.

Vocabulary · use the correct technical terms

Sign magnitude — A signed representation with a sign bit followed by magnitude bits.

Two’s complement — A signed binary representation whose most significant bit has a negative place value.

Magnitude — The size of a value without its sign.

Most significant bit — The leftmost bit in a binary representation.

Independent skill check · apply the lesson

Using 8-bit two's complement, calculate 23 + 19 and 18 - 29. Show the binary working and signed results.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Encode −13 in two representations

Follow the worked method

1. Write +13 in eight bits — CODE:
13 = 8 + 4 + 1
+13 = 00001101 — Both signed representations begin from the positive magnitude.

2. Invert, then add 1 — CODE:
00001101
invert → 11110010
add 1 → 11110011

3. Convert the two’s-complement result to denary — CODE:
−128 + 64 + 32 + 16 + 2 + 1 = −13 — Invert-and-add-one negates a value in a fixed width.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Build the answer

CLUES · uncover one at a time

Clue 1: For 8-bit two's complement, the leftmost place has weight -128.

Clue 2: Write the weights of every bit set to 1 in 11100110.

Clue 3: Add those signed place values; the final number should be negative.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: List the signed place values.

Step 2: Add the signed values.

Step 3: State the denary value and check the sign.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Two representations

Represent −37 in (a) sign magnitude [2] and (b) two’s complement [2]. Show the positive binary value and the change made for each method.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Explain an overflow
A processor adds 01110000 and 00100000 as eight-bit two’s-complement integers.
State both input values in denary. [1]
Give the stored eight-bit result. [1]
Explain whether signed overflow has occurred. Refer to the mathematical total and the representable range. [2]
State the minimum signed two’s-complement width needed to store the correct result. [1]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In 8-bit two’s complement, the signed range is −128 to +127.

Use key vocabulary from memory

1. Explain Sign magnitude in your own words.

2. Explain Two’s complement in your own words.

3. Explain Magnitude in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

11100110 = -128 + 64 + 32 + 4 + 2 = -26.

PAGE 3 · Independent skill check:

23 + 19: 00010111 + 00010011 = 00101010 (+42). 18 - 29: 00010010 + 11100011 = 11110101 (-11).

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-002/answers.html